The Impact of SME Support on Supply Chain Networks

Preliminary

Seunghoon Lee* Jee-Hyeong Park† Heewon Yang‡

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Abstract

We study the effects of government support programs for small and medium enterprises (SMEs) on supply chain networks in Korea. By linking administrative data on SME support participation and domestic firm-to-firm transactions, we construct a panel of Korean supply chain networks and document stylized facts about these networks. Using a staggered difference-in-differences design, we estimate the causal effects of SME support on manufacturing firms' network outcomes. We find that supported firms diversify their trading relationships by adding more partners, connecting to larger firms, transacting over greater geographic distances, and reducing dependence on their largest trading partner. These findings highlight an additional benefit of SME support policies through greater supply chain diversification.

JEL Classification: L14, L25, L52, H50

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*Department of Economics, Yonsei University. Email: seunghoonlee@yonsei.ac.kr
†Department of Economics, Seoul National University. Email: j-hpark@snu.ac.kr
‡Department of Economics, Yonsei University. Email: hwy1226@yonsei.ac.kr

1 Introduction

Supply chain networks play a central role in modern economies by shaping firms' access to markets and inputs, the propagation of economic shocks, and the resilience of production systems. Recent research has shown that firm performance is determined not only by productivity, but also by firms' ability to form and maintain valuable trading relationships (e.g. Bernard et al. (2022)). Because firms are embedded in production networks, policies that support firms may reshape trading relationships and generate spillovers beyond directly treated firms. A substantial literature has examined the effects of SME support on firm-level outcomes such as productivity, innovation, exports, and employment—Criscuolo et al. (2019), Bronzini and Piselli (2016), and Dechezleprêtre et al. (2023) document these effects—but much less is known about how these policies affect supply chain networks.

To address this gap, we construct a novel firm-level panel that links two administrative datasets covering the Korean economy: Small and Medium Business Integrated Management System (SIMS), which records the full history of firms' participation in government SME support programs (28.34 million support records and 4.64 million establishment-year observations, 2010–2020), and KEDVAT, VAT-based transaction data capturing the universe of domestic firm-to-firm sales. This linkage allows us to observe each firm's detailed support history alongside its complete set of domestic trading relationships, including the number, size, location, and geographic distance of its partners, as well as transaction values, over the period 2016–2024. Using this linked dataset, we study how industrial policies, such as SME support programs, affect the structure of supply chain networks for the first time in the literature.

Korea provides a particularly compelling setting for this analysis. As one of the world's leading manufacturing economies and a central node in global supply chains—spanning electronics, semiconductors, automobiles, and shipbuilding—Korea is a context where the role of industrial policy in shaping production networks is both large in scale and high in policy relevance. Our linked dataset covers the domestic economy at close to universal scope: in a 2022 snapshot, 99% of firm-to-firm transaction records are matched to SIMS, jointly spanning the full history of SME policy participation and the universe of domestic buyer–seller relationships. This combination of economic significance and data breadth makes Korea an ideal laboratory for studying whether

and how government support reshapes supply chain structure.

Firm-level outcomes such as sales, employment, and exports are often the very measures used in program screening and evaluation, making them susceptible to endogenous adjustment and strategic reporting around participation. Firms' supply chain network positions—the number, size, location, and concentration of their trading partners—are not directly targeted by any of the programs we study, and are therefore less prone to these forms of manipulation. This makes network outcomes an attractive lens through which to evaluate SME support: they are both policy-relevant and less contaminated by the endogeneity concerns that afflict conventional measures.

Nonetheless, network outcomes may still reflect endogenous selection into support programs and pre-existing differences in firms' growth trajectories. To address these concerns, we exploit the staggered timing of firms' entry into support programs and apply the difference-in-differences estimator of Callaway and Sant'Anna (2021), which is designed for settings with staggered treatment adoption and heterogeneous treatment effects. We find that SME support leads to significant expansion and diversification of firms' supply chain networks: supported firms add more trading partners, connect with larger and more geographically distant partners, and reduce dependence on their largest buyer. A central contribution of this paper is to show that SME support programs affect not only the firm-level outcomes for which they are typically designed and evaluated—sales, employment, and exports—but also the architecture of supply chain networks, reshaping who firms trade with, at what scale, and over what geographic distances.

A growing body of work uses firm-to-firm transaction data to document the structure of production networks and study how shocks propagate through them. Bernard et al. (2019) characterize buyer–supplier networks in Japan, showing that larger firms hold more trading partners, that most relationships are geographically local, and that degree assortativity between buyers and sellers is negative. Dhyne et al. (2021) use Belgian transaction data to show that the domestic production network is sparse, with a typical firm having a handful of suppliers and buyers, and that the largest buyer–supplier links are highly persistent over time. Carvalho et al. (2021) use the 2011 Great East Japan Earthquake to show that network structure shapes how localized shocks propagate to distant firms through input-output linkages. This literature has primarily treated the production network either as a static object to be described or as a conduit for the propagation of exogenous shocks, with comparatively little known about how the network itself responds to

firm-level policy interventions.

A separate literature evaluates the causal effects of SME support programs using firm-level outcomes. Criscuolo et al. (2019) estimate the effects of UK enterprise zone policies on firm employment and investment, Bronzini and Piselli (2016) study how R&D subsidies affect firm-level innovation outcomes, and Dechezleprêtre et al. (2023) use a regression discontinuity design to identify the effects of R&D tax incentives on R&D spending, patenting, and productivity spillovers. This literature is largely silent on whether SME support generates spillovers through firms' supply chain relationships.

Closest to our work, Alfaro-Ureña et al. (2022) use Costa Rican tax and transaction data to show that a government program matching domestic suppliers with multinational buyers causally affects firms' probability of forming new supply linkages and, through this channel, their subsequent performance. In their setting, however, the policy intervention is itself a network event: gaining an MNC buyer mechanically alters a firm's set of trading partners, and the resulting expansion in other relationships operates as a downstream consequence of this initial linkage. SME support programs are different in kind, as financial assistance, training, and certification target firm-level capabilities and constraints with no direct bearing on the identity of a firm's trading partners. Whether such programs nonetheless reshape a firm's network position, and through which margins, is therefore an open question rather than a near-mechanical implication of the treatment definition.

This paper bridges these literatures. Relative to the production network literature, we provide causal evidence that firm-level policy interventions—not only exogenous shocks—systematically reshape the architecture of supply chain networks, changing who firms trade with, at what scale, and over what distances; and we document new stylized facts about Korean supply chain networks using comprehensive transaction data. Relative to the SME support literature, we introduce firms' network position—the number, size, location, and concentration of their trading partners—as an outcome that is both policy-relevant and less susceptible to the endogeneity concerns affecting conventional firm-level measures, and we show that standard evaluations may underestimate the broader benefits of SME support by overlooking these network-level effects.

A further distinguishing feature of our analysis is breadth. Existing causal studies of SME and industrial policy—whether studying enterprise zones (Criscuolo et al., 2019), R&D subsidies

(Bronzini and Piselli, 2016), R&D tax incentives (Dechezleprêtre et al., 2023), or supplier-matching programs (Alfaro-Ureña et al., 2022)—each examine a single, narrowly defined instrument. We instead study the effects of a comprehensive government support system spanning seven major program categories—finance, human resources, management, startup, export, domestic sales, and technology—and ask whether network expansion and diversification are general consequences of SME support, rather than artifacts of any particular program design. By linking this full policy history with the universe of domestic firm-to-firm transactions, we provide economy-wide causal evidence on how government support reshapes supply chain structure.

The remainder of the paper proceeds as follows. Section 2 describes our two administrative datasets—SIMS and KEDVAT—and their linkage. Section 3 documents stylized facts about the structure of Korean supply chain networks. Section 4 presents cross-sectional evidence on the associations between firm size, policy support, and network outcomes. Section 5 presents the causal identification strategy and main results. Section 6 concludes.

2 Data

2.1 SIMS: SME Support Program Records

Our first dataset is SIMS (Small and Medium Business Integrated Management System), an integrated system that tracks SME support policies and support histories. SIMS records extend through the present and cover the full history of each establishment's participation in government support programs. SIMS covers seven major support program categories—Finance, Human Resources (HR), Management, Startup, Export, Domestic Sales, and Technology—along with many sub-programs.

Firms that receive government support are required to provide data covering three years before and five years after the support year. The data include establishment-level characteristics: sales, profit, employment, exports, location, industry, and other business attributes, as well as the amount, date, and program type of each support event. SIMS is not limited to SMEs; it also includes additional data from KoDATA on listed firms and credit-rated firms, providing broader coverage of the firm population.

As of a 2010–2020 snapshot, SIMS contained 28.34 million government support records and

4.64 million establishment-year observations. In a 2019 cross-section, 68% of the 0.65 million establishments in SIMS received financial support (approximately 0.45 million establishments), while 23% received HR support. The remaining program categories each account for roughly 3–11% of establishments; since firms commonly participate in multiple programs, these shares sum to more than 100%. Over an eight-year window (three years pre- and five years post-support), most SMEs receive at least one form of government support.

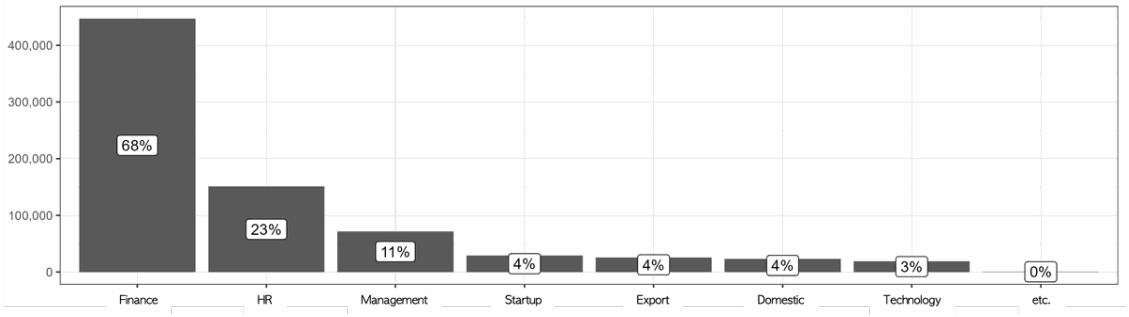


Figure 1: Number of Establishments by Support Program (2019)

Notes: Percentages do not sum to 100% because establishments may receive support from multiple program categories in the same year.

2.2 KEDVAT: Firm-to-Firm Transaction Data

The second dataset is KEDVAT, a VAT-based firm-to-firm transaction database covering 2016–2024. KEDVAT contains roughly 2 billion transaction records and 29 million firm-year observations (i.e., firms are repeat-counted across the nine years of coverage). The unit of observation is a firm-to-firm transaction by reporting period (quarterly, semi-annual, and annual). For example, reporting firm A trades with partner firm B 100 times with a total trade value of 100 million KRW in 2022Q1.

Each transaction record is classified as either a *sale* (the reporting firm is the seller) or a *purchase* (the reporting firm is the buyer), allowing us to construct separate sales- and purchase-side network measures for each firm. KEDVAT includes pseudonymized buyer and seller identifiers that are linkable to SIMS, transaction amounts, VAT amounts, and industry classification codes. This allows us to observe domestic trading relationships at the buyer–seller level over time.

While KEDVAT records extend through 2024, the sample window available for any individual

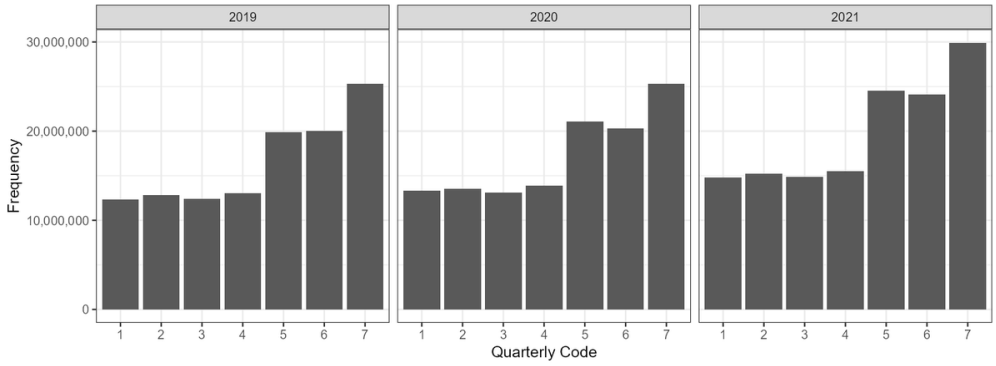


Figure 2: Observation Counts by Period Code (2019–2021)

Notes: Period codes 1–4 correspond to quarterly filings, codes 5–6 to semi-annual filings, and code 7 to annual filings. The years 2019–2021 are shown as representative of the overall pattern. The higher frequency of codes 5–7 indicates that most firms file on a semi-annual or annual basis rather than quarterly.

firm reflects a rolling three-year agreement: firms that consented to share data for credit-rating purposes provide records for the three years before and after the reference year. As a result, coverage becomes progressively less complete toward the most recent years, and Figure 3 shows a sharp decline in transaction counts after 2022. This decline reflects partial coverage from these rolling agreements rather than an underlying drop in economic activity. We therefore restrict our main analysis sample to 2016–2022, the period for which firm coverage is most complete.



Figure 3: Annual Transaction Counts, by Transaction Type

2.3 Linking SIMS and KEDVAT

With the cooperation of the Korea Small Business Institute (KOSI), we link the two datasets using pseudonymized identifiers based on business registration numbers. In a 2022 snapshot, 99% of KEDVAT transaction records are matched to SIMS for at least one side of the transaction: 71% are matched on both sides, 27% only for the reporting firm, and 1% only for the partner firm. Among reporting firms, 91% are matched to SIMS; among partner firms, 41% are matched.

For partner firms not in SIMS, we retain industry codes and unique identifiers from KEDVAT based on their business registration numbers. All empirical analyses are conducted at the establishment level, and for expositional convenience we use "firm" and "establishment" interchangeably unless otherwise specified.

2.4 Summary Statistics

The Korean economy has several distinctive structural features that shape the network patterns we document below. SMEs are pervasive in the Korean economy: over 99 percent of establishments qualify as SMEs by employee count, and more than half have fewer than five employees. Government support is correspondingly extensive, delivered through programs spanning finance, human resources, management, startup, export, domestic sales, and technology, such that the majority of SMEs receive at least one form of support over a multi-year window. Korean production is organized hierarchically around large business groups (chaebol), with SMEs commonly engaged in subcontracting relationships and heavily dependent on a small number of large buyers. Economic activity is also highly spatially concentrated: the Seoul Metropolitan Area (SMA) covers only 11.8 percent of national territory but contains 50.6 percent of the population and accounts for 52.7 percent of gross regional domestic product (OECD, 2025), and reducing this concentration has long been a central objective of Korean regional development policy. These features—the dominance of SMEs, the breadth of support programs, the hierarchical production structure, and the geographic concentration of activity—provide the backdrop for the network patterns we document below.

Table 1 shows economy-wide summary statistics for the 2022 sample, covering establishments across all industries. The variables capture multiple dimensions of firms' supply chain networks: scale (number of partners, average transaction volume), quality (employment size of top and bot-

tom 10% partners), structure (top partner share, geographic distance, local share), and the share of transactions within the Seoul Metropolitan Area.

Support Status is an indicator equal to one if the establishment received support from any of the seven SIMS program categories at any point during the sample period; it is not specific to any single program type. The treatment variable used in the staggered difference-in-differences analysis (Section 5.1) is defined separately, based on the first receipt of HR support. Several network variables, most notably the number of partners, exhibit highly right-skewed distributions (Table 1). In the regression analyses below, these variables are log-transformed.

Table 1: Summary Statistics (2022, All Industries)

Variable	Mean	SD	Min	Max	N
Number of Partners	70.662	652.338	1.000	96,411.000	207,145
Average Transaction Volume (Million USD)	0.083	0.494	0.000	83.939	206,927
Avg. Top 10% Partners' Emp. Size	1,533.413	6,277.588	0.000	120,933.000	201,605
Avg. Bottom 10% Partners' Emp. Size	600.602	3,623.806	0.000	120,933.000	201,605
Avg. Partners' Emp. Size	807.149	2,950.130	0.000	120,933.000	201,605
Average Distance to Partners (m)	67,119.414	56,732.147	0.000	489,957.445	196,690
Top Partner Share	0.464	0.287	0.000	1.000	207,145
SMA Share	0.502	0.374	0.000	1.000	196,690
Local Share	0.400	0.334	0.000	1.000	196,690
Employment Size	24.663	195.041	0.000	35,343.000	186,605
Support Status	0.436	0.496	0.000	1.000	207,145

Notes: Average transaction volume is reported in million USD, using an exchange rate of 1 USD = 1,400 KRW. Support Status equals one if the establishment received support from any SME support program (across all seven categories) during the sample period; see text for the treatment definition used in the difference-in-differences analysis.

3 Stylized Facts on Korean Supply Chains

3.1 Customer Concentration

We document several stylized facts about Korean supply chain networks. Throughout this section, for simplicity, we classify establishments with ≤ 300 employees as SMEs across all industries. We measure *top partner dependence* as the share of an establishment's total sales (or purchases) accounted for by its single largest trading partner.[1] SMEs (≤ 300 employees) rely heavily on a small number of customers, often a single top buyer, on the sales side, while maintaining more

[1]This differs from the Top Partner Share variable used in the regression analysis (Table 1), which is defined as the average share accounted for by an establishment's top 10% of partners, ranked by transaction value.

diversified supplier bases on the purchase side. Figure 4 shows this pattern by firm size, and Figure 5 shows similar patterns across industries. In most industries except wholesale & retail, many firms exhibit extreme dependence (90–100% of sales) on a single buyer. On the purchase side, extreme dependence is uncommon except in business support & rental.

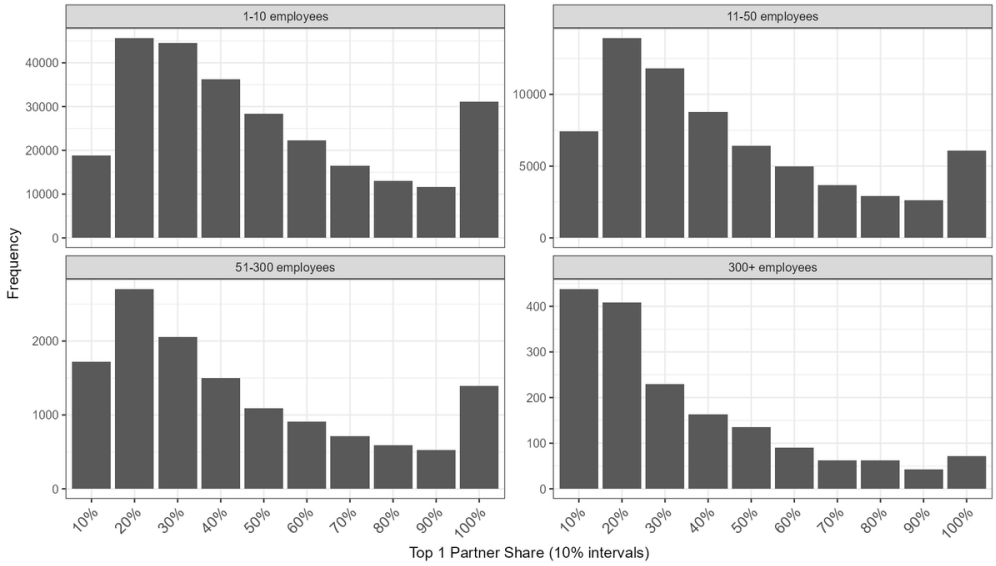


Figure 4: Top Partner Dependence in Sales by Firm Size (2022)

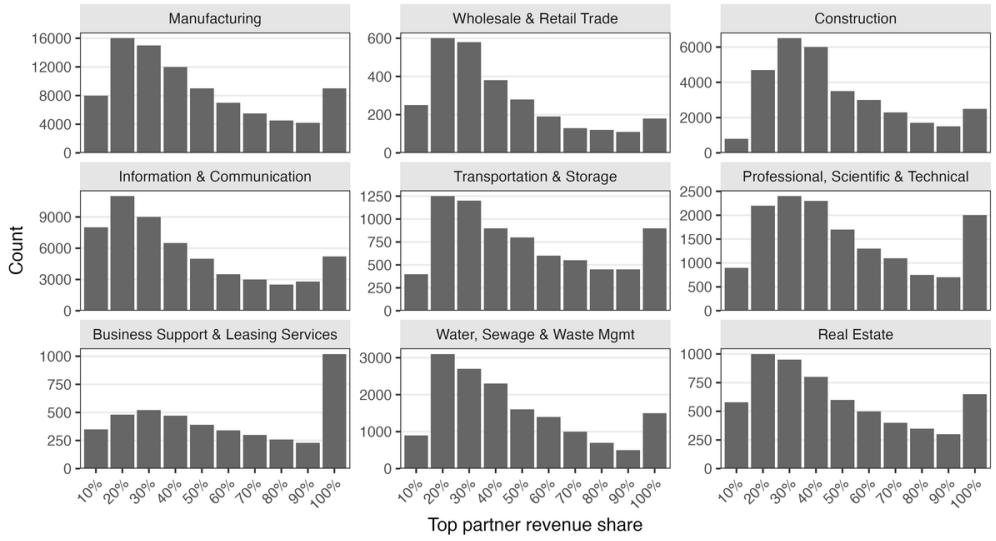


Figure 5: Top Partner Dependence in Sales by Industry (2022)

3.2 Average Number of Partners

The average number of trading partners differs substantially across industries and between sales and purchase networks (Figure 6). Service and public, energy sectors exhibit a larger number of partners on the sales side than manufacturing, whereas the opposite pattern appears on the purchase side. The average number of partners in the public, energy sector rises sharply in 2021, possibly reflecting either policy responses to COVID-19 or changes in sample composition; we do not pursue this issue further here.

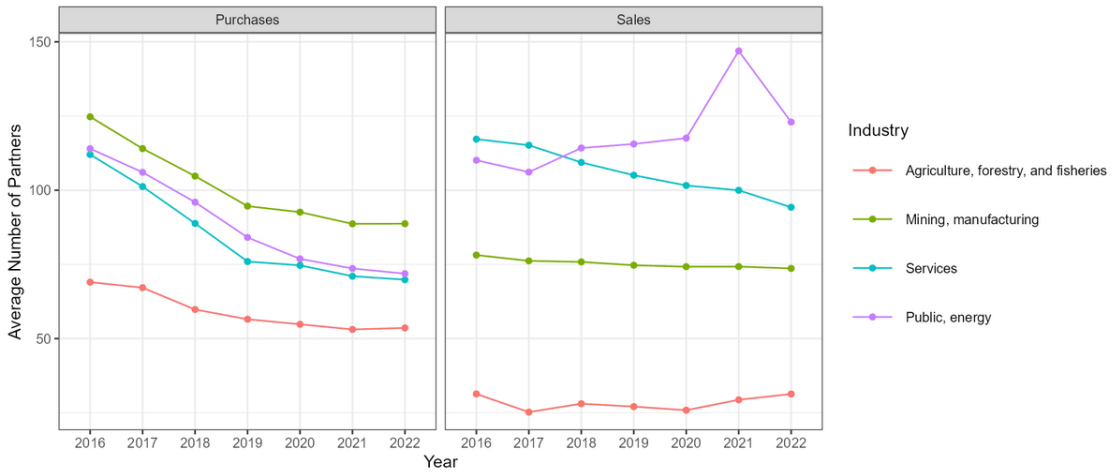


Figure 6: Average Number of Partners by Industry

Taken together, these descriptive patterns suggest that customer concentration and network diversification are economically important dimensions of Korean firm networks, especially for SMEs.[2] This motivates our policy analysis below, which examines whether SME support helps firms expand and diversify their transaction relationships.

4 Cross-Sectional Analysis

We first examine the cross-sectional associations between firm size, SME support receipt, and network outcomes. Although these associations are not causal, they are informative for two reasons. First, the relationship between firm size and supply chain network structure is theoretically ambiguous. Larger firms might hold more diversified trading relationships and transact

[2] As a check on the representativeness of KEDVAT, we verify that Korean firm networks exhibit the core regularities documented for Japan by Bernard et al. (2019)—larger firms have more trading partners, most buyer–supplier links are local, and seller–buyer degree assortativity is negative; see Appendix A.

with more geographically distant partners if scale expands the set of profitable matches; but they might equally concentrate transactions among a smaller number of large, closely located partners if relationship-specific investments generate increasing returns to concentrated ties, or if proximity lowers coordination costs. The direction of these associations is therefore an open empirical question. Second, these patterns have not been systematically documented in an economy with Korea's distinctive industrial structure—organized around large business groups, with economic activity heavily concentrated in the Seoul Metropolitan Area, and with a manufacturing sector that plays an outsized role in overall output. Whether the associations between firm size, policy support, and network structure observed in other countries carry over to this setting is not self-evident, and establishing them is a contribution in its own right.

4.1 Specification

We examine cross-sectional associations between SME support and network outcomes using the following specification:

$$y_{i,s,d} = \beta_1 X_i + \beta_2 U_i + \delta_s + \delta_d + \varepsilon_{i,s,d} \tag{1}$$

where $y_{i,s,d}$ is a network outcome for establishment i in industry s and region d; X_i is log employment size; U_i is an indicator equal to one if establishment i received any SME policy support in the previous quarter (Q2 2022); and δ_s, δ_d are industry (KSIC) and region fixed effects. The sample is a cross-sectional snapshot of manufacturing firms in Q3 2022.

We consider nine dependent variables capturing network scale, quality, and structure: the number of trading partners; the average transaction volume per partner; the average employment size of partners in the top and bottom deciles (by transaction value); the overall average employment size of partners; the share of transactions accounted for by the largest partner; the average distance to partners; the share of transactions within the Seoul Metropolitan Area; and the share of transactions within the same region. These correspond to the variables summarized in Table 1.

We do not interpret the estimates from (1) causally. Because policy receipt is non-random, cross-sectional differences may reflect selection into support programs rather than the effects of support itself.

4.2 Results

Figure 7 reports estimates from the cross-sectional specification (1). The coefficients on log employment size are consistent with the stylized facts documented in Section 3: larger firms have more trading partners, transact with larger partners, and transact over greater distances.

The coefficients on the support indicator point to a different pattern. Controlling for firm size, industry, and region, firms receiving support have more trading partners and transact over greater distances (similar to the size effect), but they also exhibit lower average transaction volumes per partner and transact with smaller partners on average, the opposite of the size effect. In other words, supported firms do not simply resemble larger firms along all dimensions; rather, support is associated with a distinct pattern of network expansion characterized by a larger number of smaller-scale trading relationships. Results on the purchase side are qualitatively similar.

We interpret these patterns descriptively rather than causally. This contrast is suggestive but cannot distinguish between two competing explanations. On one hand, support may causally induce firms to expand their networks through a larger number of smaller-scale relationships. On the other hand, firms that already have smaller, more numerous partners (perhaps because they are less productive or less established) may simply be more likely to receive support in the first place, in which case the cross-sectional association would reflect selection rather than a treatment effect. Distinguishing between these explanations requires causal identification, to which we now turn.

5 Causal Analysis

5.1 Empirical Strategy

For causal identification, we apply the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), which addresses heterogeneous treatment timing by estimating group-time average treatment effects and aggregating them to an overall ATT. We focus primarily on HR support, the second-largest component of SME support programs after financial support, because financial programs are highly heterogeneous in design (e.g., credit guarantees, direct loans), which

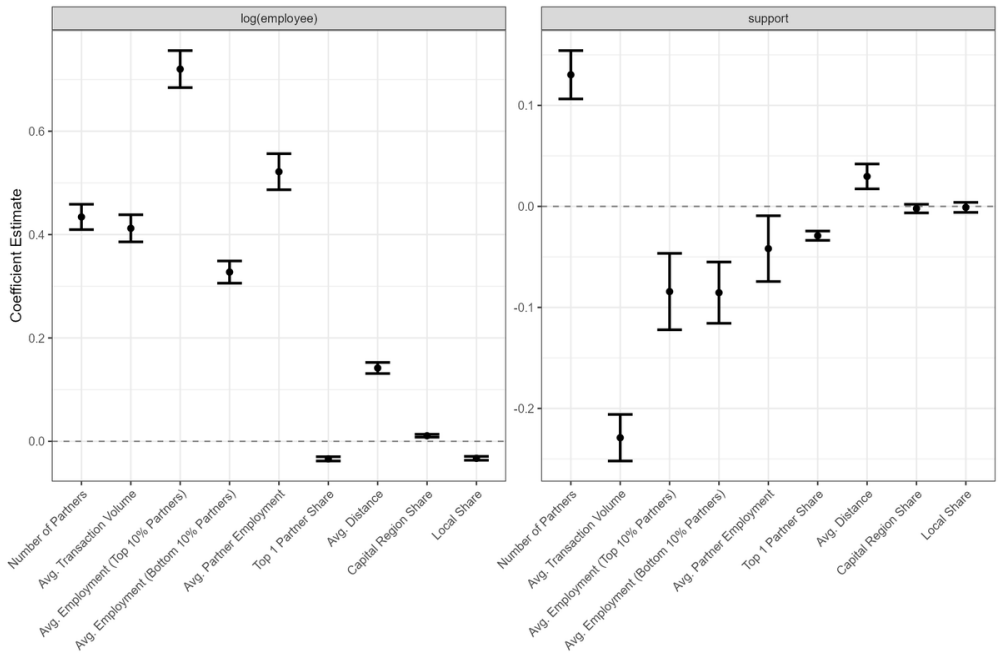


Figure 7: Cross-Sectional Analysis Results (Sales)

complicates the interpretation of treatment effects.[3] The treatment group consists of manufacturing firms that first received HR support between Q1 2017 and Q2 2019. Firms first treated in Q3 2019 or later are excluded from the analysis sample entirely; the control group therefore consists of firms within the 2016–2022 window that either had not yet received HR support by the end of the sample period (not-yet-treated) or never received it (never-treated).

This sample restriction is motivated by a shift in the nature of HR support programs around the onset of the COVID-19 pandemic. As shown in Appendix Figure 16, HR support disbursed between 2017 and the first half of 2019 was concentrated in growth-oriented sub-programs—such as general workforce hiring and working environment improvement—aimed at expanding firms' workforce and operational capacity. From the second half of 2019 onward, HR support shifted sharply toward defensive, pandemic-related programs such as employment retention subsidies, which are oriented toward preserving existing employment and trading relationships rather than expanding them. Including firms first treated under this later, qualitatively different policy regime would introduce substantial heterogeneity in the treatment definition itself. We apply the same sample restriction when estimating analogous specifications for financial and export support in

[3]Figure 8 shows the number of manufacturing establishments receiving support, by program type and year.

Section 5.3.

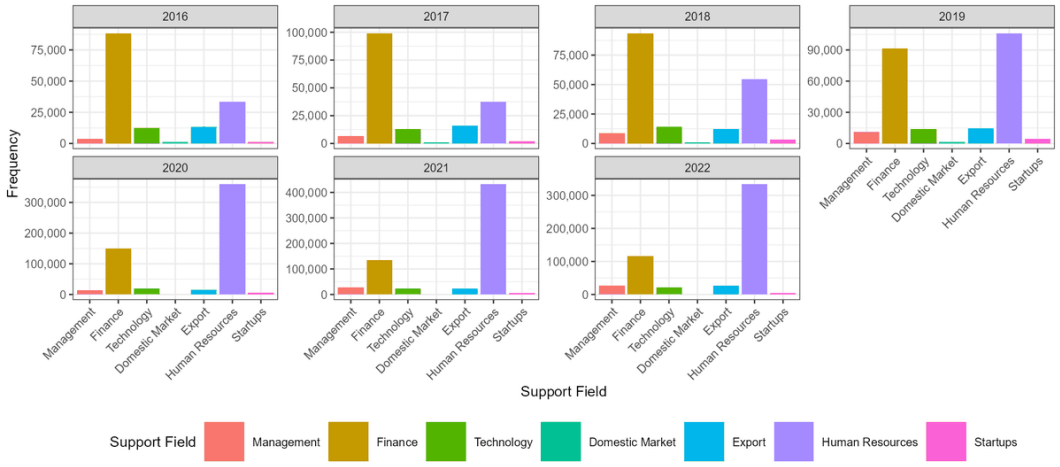


Figure 8: Number of Support Cases by Program Type for Manufacturing Establishments

The key identifying assumption is conditional parallel trends: in the absence of treatment, the evolution of network outcomes for treated firms would have followed the same path as for control firms, conditional on covariates. We control for log sales and 2-digit KSIC industry, entered as a set of industry indicators, and report event study plots to assess pre-trend violations. We estimate effects using the doubly robust (DR) estimator, which combines outcome regression with inverse propensity score weighting and is consistent if either model is correctly specified; covariates enter both components as described above.[4]

5.2 Causal Effects of HR Support

We estimate the staggered difference-in-differences specification described in Section 5.1 for the sales side of firms' supply chain networks. Given that the cross-sectional patterns documented above are qualitatively similar on the purchase side, we focus the causal analysis on sales-side outcomes. Figure 9 presents event study estimates for eight network outcomes. Across all outcomes, pre-treatment coefficients are statistically indistinguishable from zero, supporting the conditional parallel trends assumption.

The results show that HR support expands and diversifies firms' supply chain networks along

[4]In the current specification, 2-digit KSIC industry codes enter as a continuous covariate. We are re-estimating the model with industry entered as a full set of categorical indicators; preliminary checks using log sales alone yield qualitatively similar results, but we will update the reported estimates accordingly.

several dimensions. In terms of scale, supported firms continuously acquire new trading partners following the receipt of HR support (Figure 9a). This continuous accumulation of new partners, rather than a one-time jump, suggests that HR support operates as an ongoing capability-building intervention rather than a transitory shock to firms' transaction relationships. The corresponding decline in average transaction volume per partner is consistent with this rapid partner expansion: as firms add new trading relationships, average volume per partner mechanically falls unless total transaction volume increases proportionally (Figure 9b).

HR support also enhances firms' ability to connect with larger partners at the upper end of the partner distribution. Average employment size among the top 10% of partners increases significantly post-treatment (Figure 9c), while no substantial changes are observed among bottom-10% partners (Figure 9d).[5]

This finding resolves the ambiguity raised in Section 4.2. In the cross-section, supported firms exhibit smaller average partner size than comparable non-recipients, consistent with negative selection, whereby less-connected firms disproportionately seek and receive government assistance. Yet the causal estimates tell a strikingly different story: following treatment, these same firms achieve significant upward upgrading in partner quality at the top of the distribution. Support recipients are thus negatively selected on network quality at baseline, yet they experience positive network upgrading after treatment. The cross-sectional pattern of smaller average partner size therefore reflects a pre-existing disadvantage of recipients rather than a causal consequence of support.

Firms also exhibit a persistent increase in the average geographic distance to their partners (Figure 9e), while sales dependence on the top trading partner declines (Figure 9f). The decline in top-partner dependence is particularly notable in light of the customer concentration documented in Section 3, where a substantial share of SMEs were found to rely on a single buyer for the majority of their sales; HR support appears to help firms move away from this pattern of concentrated dependence.

Despite these increases in the number of partners and the average distance to partners, the share of transactions within the Seoul Metropolitan Area and within the local region does not

[5]We omit results for the overall average employment size of partners, as this measure is closely related to—and largely driven by—the top- and bottom-decile measures shown here.

decline significantly (Figures 9g–9h). This indicates that firms expand their networks through new trading relationships while continuing to maintain existing regional ties, rather than substituting away from local or capital-region partners. Taken together, these results suggest that HR support does not simply reallocate a fixed set of trading relationships, but instead enables firms to add new relationships (with larger, more distant partners) on top of their existing network.

These findings are consistent with several possible mechanisms. SME support may reduce the costs of establishing new business relationships through networking opportunities, consulting services, or information provision. Alternatively, support may relax resource constraints and enable firms to expand production capacity, making them more attractive trading partners.

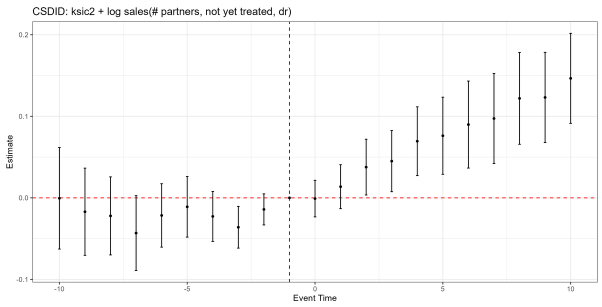
5.3 Heterogeneity by Policy Type

We also estimate analogous specifications for financial and export support policies and compare the results. Figure 10 reports the overall ATT for four outcomes—average distance, top-partner share, SMA share, and local share—across the three policy types. All three policy types have positive effects on geographic and structural diversification, although the magnitudes differ.

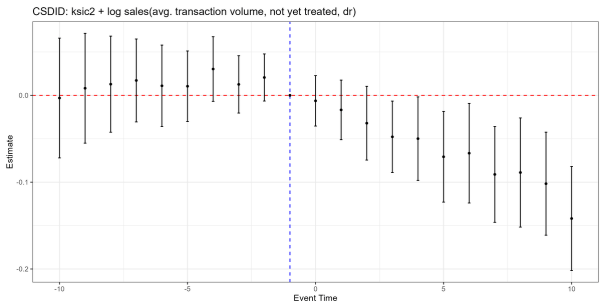
Financial support exhibits larger effects than HR and export support for average distance, top-partner share, and local share. One possible explanation is that financial support more directly relaxes firms' budget constraints, enabling the costs associated with establishing new and more distant trading relationships. Financial support programs are also substantially larger in scale than HR or export programs, which may further contribute to their larger estimated effects. The Seoul Metropolitan Area share is not significantly affected under any of the three policy types. These comparisons suggest that different policy instruments affect network expansion through distinct channels and with different intensities.

6 Conclusion

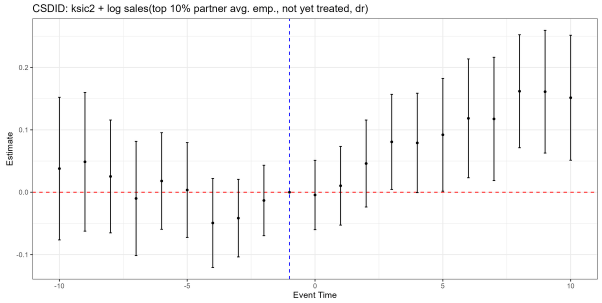
This paper studies the impact of SME support policies on supply chain network structure in Korea. Combining SIMS support records with KEDVAT firm-to-firm transaction data, we show that supported firms are negatively selected on network quality at baseline, exhibiting smaller average partner size than comparable non-recipients, yet experience significant positive network upgrad-



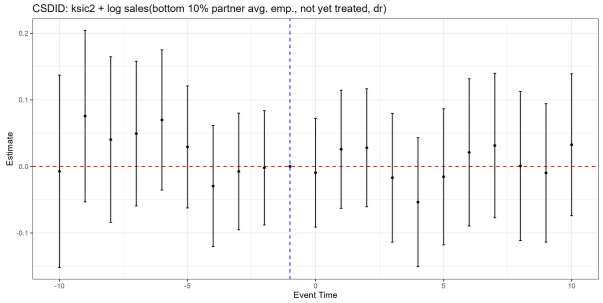
(a) Number of Partners



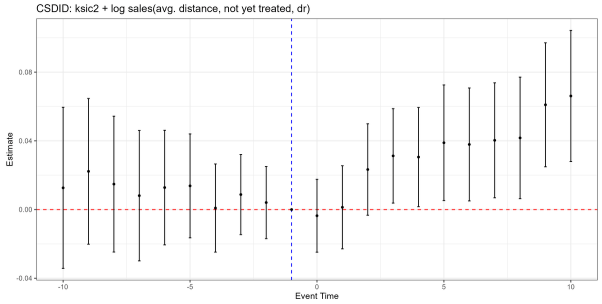
(b) Average Transaction Volume



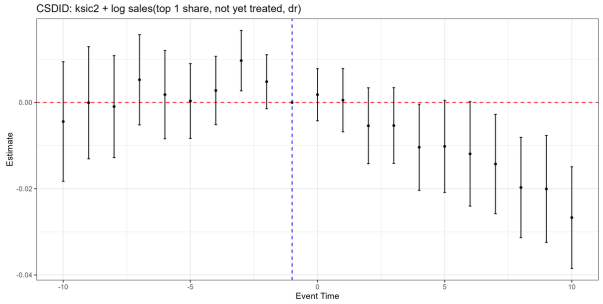
(c) Average Top 10% Partners' Employment Size



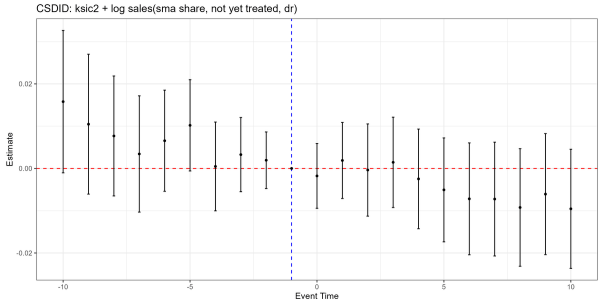
(d) Average Bottom 10% Partners' Employment Size



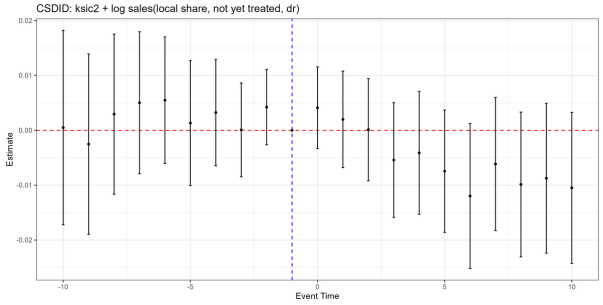
(e) Average Distance to Partners



(f) Top Partner Share



(g) Seoul Metropolitan Area Share



(h) Local Share

Figure 9: Event-Study Results for HR Support

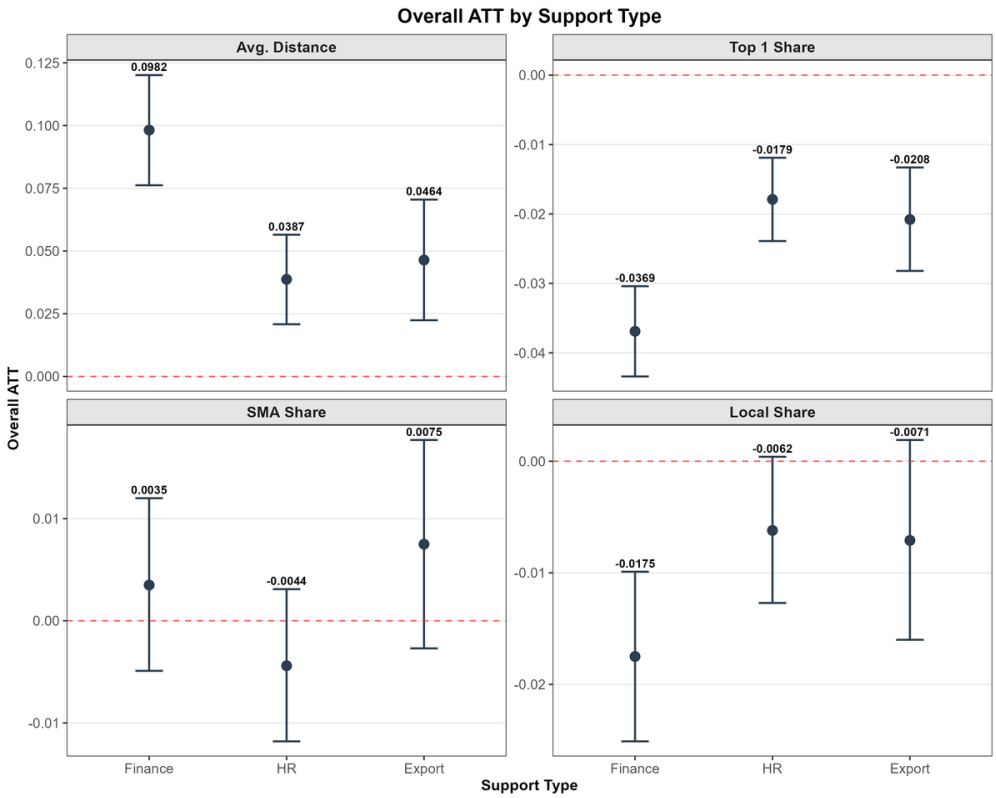


Figure 10: Overall ATT by Policy Type

ing after treatment. Staggered difference-in-differences estimates show that HR support causally increases the number of trading partners, enables firms to connect with larger partners at the top of the partner distribution, extends the geographic reach of trading relationships, and reduces dependence on the largest buyer. Financial, HR, and export support all promote network diversification, though financial support exhibits larger effects; one possible explanation is that financial support more directly relaxes firms' budget constraints.

These findings reveal an unintended benefit of SME support that is invisible to conventional evaluation frameworks. Standard assessments of SME programs focus on firm-level outcomes—sales, employment, exports, and patents—that are often the very criteria used in program design and screening. Supply chain network diversification, by contrast, is not an explicit target of any of the programs we study, yet it emerges as a systematic consequence of support. This raises the possibility that existing evaluations underestimate the total social return to SME support, since they fail to account for network-level upgrading experienced by recipients and the potential spillovers to their trading partners.

The geographic results are directly relevant to Korea's longstanding policy concern with regional economic concentration. Supported firms transact over significantly greater distances following treatment, indicating that support expands firms' geographic reach beyond their immediate region. Yet the share of transactions within the Seoul Metropolitan Area remains broadly stable post-treatment, suggesting that SME support enables firms to form new relationships in more distant regions without displacing existing ties to the capital. The implication for regional policy is nuanced: SME support promotes geographic diversification in the sense of expanding national reach, but it appears to complement rather than substitute for dedicated regional concentration policies.

Given the high customer concentration documented in Section 3, the reduction in top-partner dependence has direct implications for supply chain resilience. The COVID-19 pandemic and subsequent supply chain disruptions highlighted the systemic risks associated with concentrated trading relationships. Our findings suggest that SME support programs, even when not explicitly designed to enhance resilience, can strengthen supply chains by enabling firms to diversify their customer base and reduce dependence on any single trading partner. More broadly, they point to an additional rationale for government support of SMEs: fostering supply chain diversification and, in turn, greater resilience.

Three directions for future work are worth noting. First, we plan to estimate policy effects using a continuous treatment framework based on the amount of support received, allowing us to assess dose-response relationships. Second, we aim to quantify policy efficiency by measuring network expansion and diversification per unit of cost, which would facilitate cost-benefit comparisons across program types. Third, we plan to exploit the detailed program information in SIMS to examine heterogeneity across support categories and better understand the mechanisms through which SME support reshapes supply chain networks.

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A Appendix: Comparison with Bernard et al. (2019)

This appendix provides supporting evidence for the comparison with Bernard et al. (2019) discussed in Section 3. We examine whether Korean firm-to-firm supply chain networks, as recorded in KEDVAT, exhibit the same broad regularities documented for Japan: (i) larger firms have more trading partners; (ii) larger firms' partners are more geographically dispersed; (iii) most buyer–supplier links are local; and (iv) seller–buyer degree assortativity is negative.

Figure 11 plots the number of trading partners (both customers and suppliers) against firm sales, on a log-log scale. The relationship is strongly increasing across the size distribution, consistent with Bernard et al. (2019).

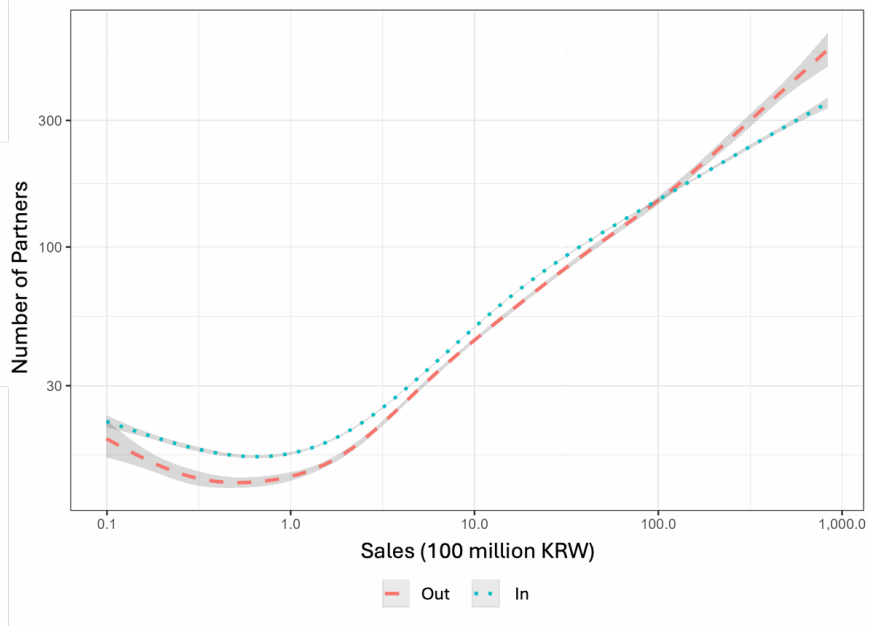


Figure 11: Number of Partners by Firm Sales (2022)

Figure 12 shows the number of distinct municipalities (si/gun/gu) in which a firm's partners are located, by firm sales. Larger firms' partners are spread across a greater number of regions, indicating that larger firms maintain more geographically dispersed networks.

Figure 13 shows the distribution of distances between firms and their trading partners. The distribution is heavily concentrated at short distances, with the majority of relationships occurring within roughly 100 km, consistent with the finding that most buyer–supplier links are local.

Figure 14 plots, for each firm, its number of suppliers (in-degree) against the average number

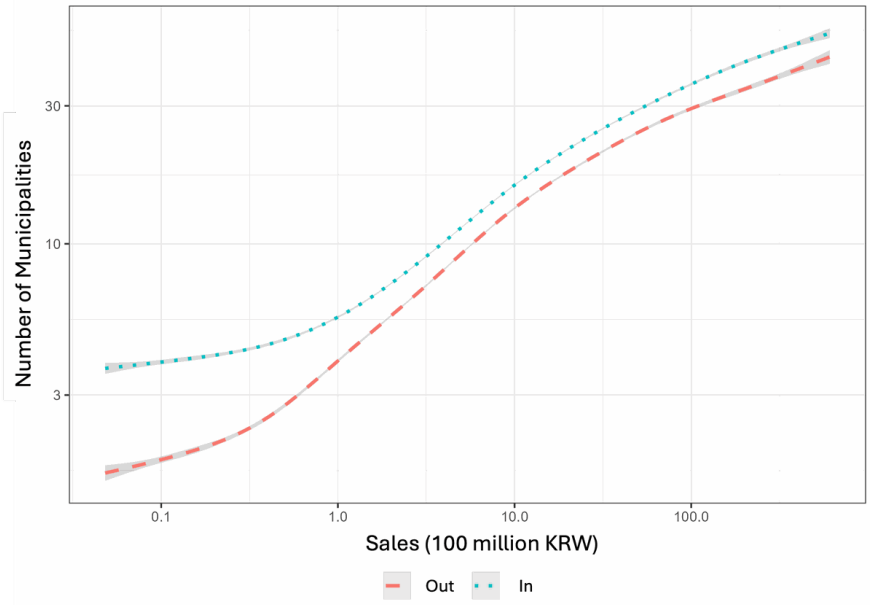


Figure 12: Number of Distinct Partner Municipalities by Firm Sales (2022)

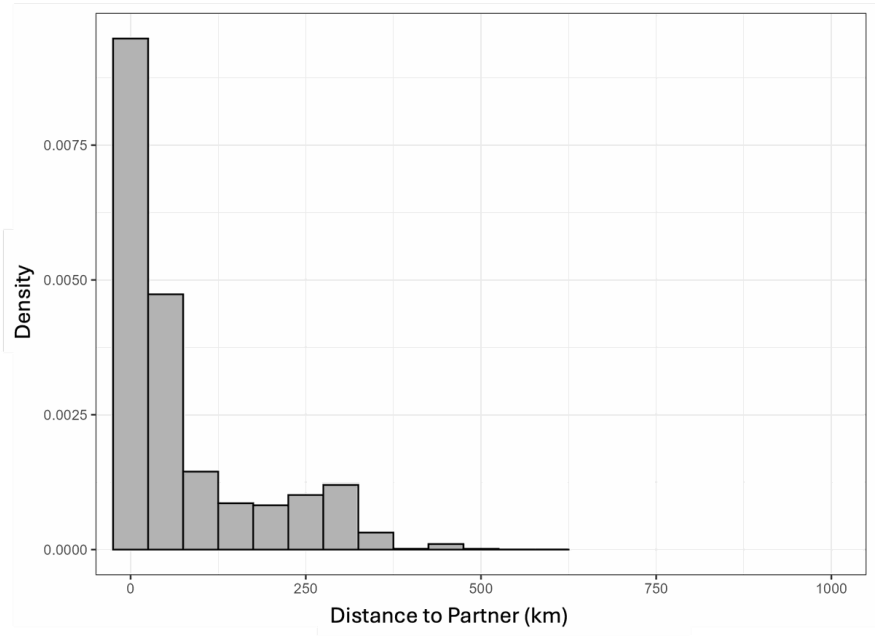


Figure 13: Distribution of Distance to Trading Partners (2022)

of customers (out-degree) of its suppliers, on a log-log scale. The fitted relationship has a slope of -0.8, indicating negative degree assortativity: firms with many suppliers tend to be supplied by firms that themselves have relatively few customers. This mirrors the negative assortativity documented by Bernard et al. (2019) for Japan.

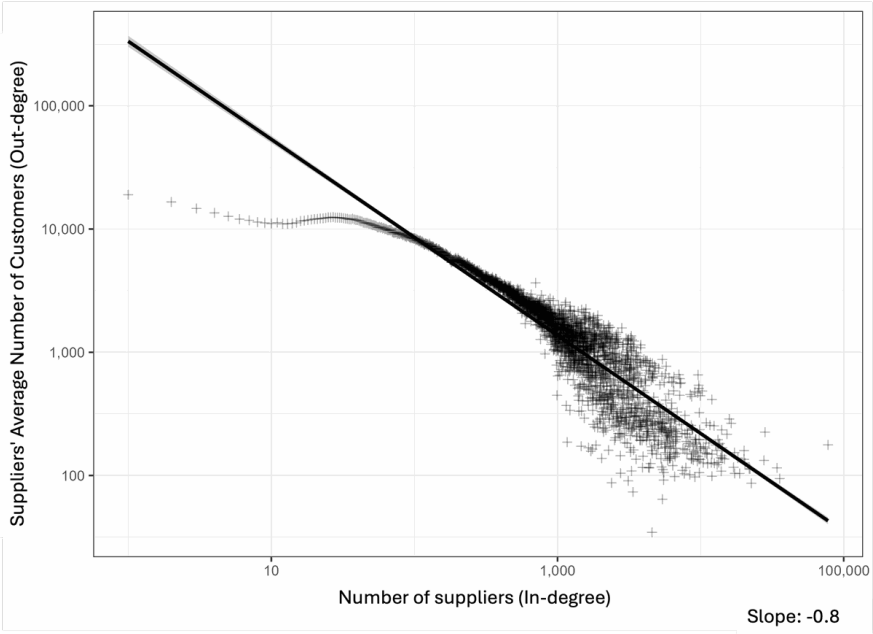


Figure 14: Degree Assortativity: Suppliers' Average Out-degree by Firm In-degree (2022)

Figure 15 shows the complementary cumulative distribution of in- and out-degree (number of suppliers and customers) on a log-log scale. Both distributions exhibit heavy right tails, with out-degree somewhat more skewed than in-degree, indicating that a small number of firms account for a disproportionate share of customer relationships.

Taken together, Figures 11–15 confirm that Korean production networks exhibit the core regularities documented by Bernard et al. (2019) for Japan, supporting the representativeness of KED-VAT as a source of firm-to-firm transaction data.

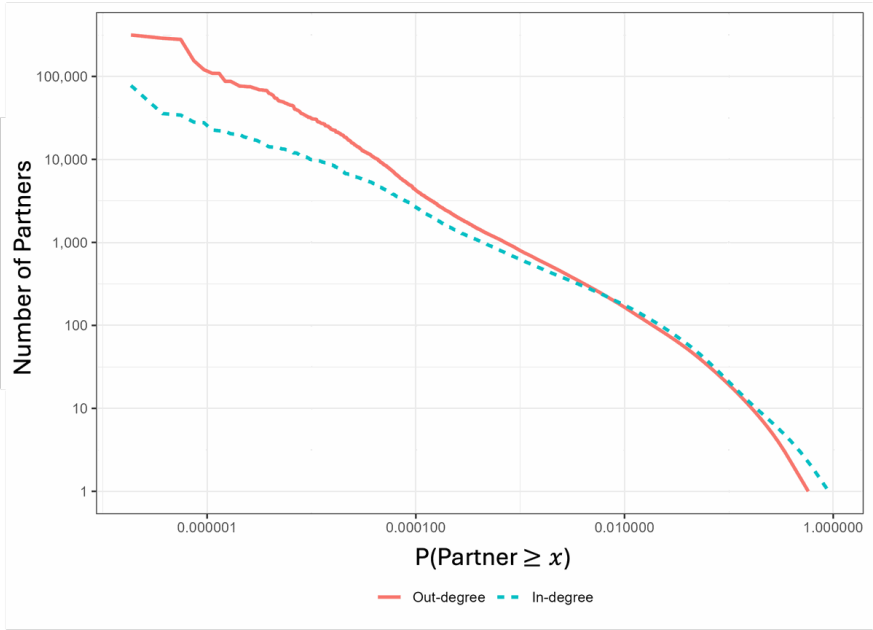


Figure 15: Degree Distribution (Complementary CDF, 2022)

B HR Support Program Details

Figure 16 shows the composition of HR support programs over time. Table 2 describes the sub-program categories used in the analysis.

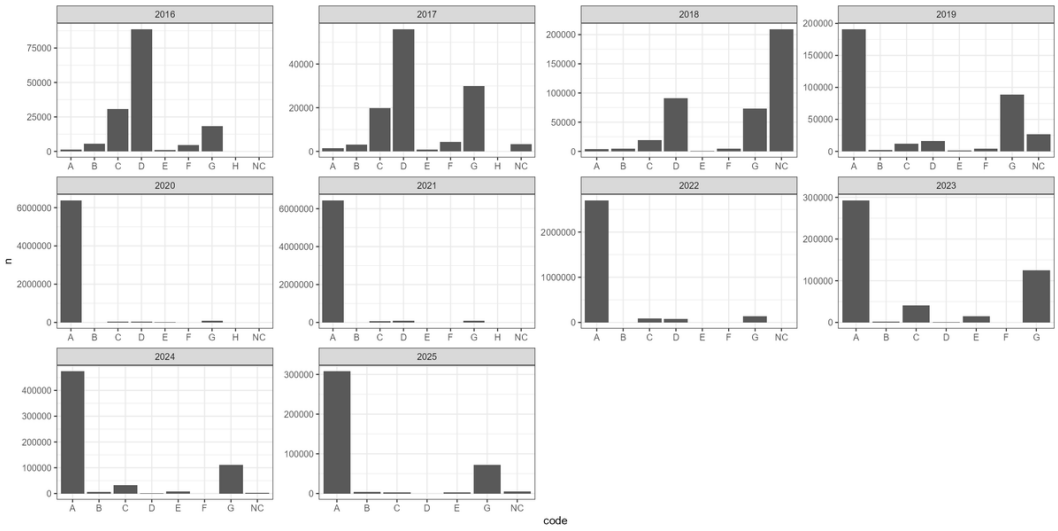


Figure 16: HR Support Program Composition Over Time

Table 2: HR Support Sub-Program Classification

Code	Description
A	Employment retention
B	Employment environment improvement
C	Education / training / workshop
D	Domestic general workforce
E	Domestic professional workforce
F	Workforce information provision
G	Working environment improvement
H	Overseas workforce programs
NC	Not classified